

Let $\Omega \subset \mathbb{R}^d$ be a domain with Lipschitz boundary, let $H_0^1(\Omega)$ be equipped with the norm

$$\|u\|_{1,\Omega} = \sqrt{(\nabla u, \nabla u) + (u, u)},$$

and let $a : H_0^1(\Omega) \times H_0^1(\Omega) \rightarrow \mathbb{R}$ be bilinear. Assume there exist constants $C, c > 0$ such that for all $u, v \in H_0^1(\Omega)$ we have boundedness

$$|a(u, v)| \leq C\|u\|_{1,\Omega}\|v\|_{1,\Omega}$$

and coercivity

$$a(u, u) \geq c\|u\|_{1,\Omega}^2.$$

Let $F \in H^{-1}(\Omega)$. Consider the continuous variational problem: find $y \in H_0^1(\Omega)$ such that

$$a(y, v) = F(v) \quad \text{for all } v \in H_0^1(\Omega).$$

Define the discrete subspace $V_h^0 \subset H_0^1(\Omega)$, assumed finite dimensional, and the discrete variational problem: find $y_h \in V_h^0$ such that

$$a(y_h, v_h) = F(v_h) \quad \text{for all } v_h \in V_h^0.$$

By the Lax–Milgram Lemma, since a is bounded and coercive on the Hilbert space $H_0^1(\Omega)$, there exists a unique $y \in H_0^1(\Omega)$ satisfying

$$a(y, v) = F(v) \quad \text{for all } v \in H_0^1(\Omega).$$

Likewise, restricting the same bilinear form a to the subspace V_h^0 preserves boundedness and coercivity on V_h^0 : for all $u, v \in V_h^0$ we still have

$$|a(u, v)| \leq C\|u\|_{1,\Omega}\|v\|_{1,\Omega}$$

and

$$a(u, u) \geq c\|u\|_{1,\Omega}^2.$$

Hence, applying Lax–Milgram again on the Hilbert space V_h^0 (with the inherited norm), there exists a unique $y_h \in V_h^0$ such that

$$a(y_h, v_h) = F(v_h) \quad \text{for all } v_h \in V_h^0.$$

Now set the error $e := y - y_h$. Subtracting the continuous identity from the discrete one yields Galerkin orthogonality. Indeed, for any $v_h \in V_h^0$ we have $a(y, v_h) = F(v_h)$ and $a(y_h, v_h) = F(v_h)$, so

$$a(y - y_h, v_h) = 0,$$

i.e. $a(e, v_h) = 0$ for all $v_h \in V_h^0$.

Fix an arbitrary $v_h \in V_h^0$ and define $w := y_h - v_h$, so $w \in V_h^0$. Using the identity

$$e = y - y_h = (y - v_h) - (y_h - v_h) = (y - v_h) - w,$$

and bilinearity of a , we obtain

$$a(e, e) = a(e, (y - v_h) - w) = a(e, y - v_h) - a(e, w).$$

By Galerkin orthogonality with the test function $w \in V_h^0$, we have $a(e, w) = 0$, and therefore

$$a(e, e) = a(e, y - v_h).$$

Using coercivity on the left-hand side gives

$$a(e, e) \geq c\|e\|_{1,\Omega}^2.$$

Using boundedness on the right-hand side gives

$$|a(e, y - v_h)| \leq C\|e\|_{1,\Omega}\|y - v_h\|_{1,\Omega}.$$

Combining these inequalities and the identity $a(e, e) = a(e, y - v_h)$, we get

$$c\|e\|_{1,\Omega}^2 \leq C\|e\|_{1,\Omega}\|y - v_h\|_{1,\Omega}.$$

If $\|e\|_{1,\Omega} = 0$, then the desired estimate holds trivially. Otherwise we can divide by $\|e\|_{1,\Omega} > 0$ to obtain

$$\|y - y_h\|_{1,\Omega} = \|e\|_{1,\Omega} \leq (C/c)\|y - v_h\|_{1,\Omega}.$$

Since $v_h \in V_h^0$ was arbitrary, taking the infimum over all $v_h \in V_h^0$ yields

$$\|y - y_h\|_{1,\Omega} \leq (C/c) \inf_{v_h \in V_h^0} \|y - v_h\|_{1,\Omega}.$$

Thus the continuous and discrete problems have unique solutions $y \in H_0^1(\Omega)$ and $y_h \in V_h^0$, and Cea's Lemma holds with the estimate

$$\|y - y_h\|_{1,\Omega} \leq (C/c) \inf_{v_h \in V_h^0} \|y - v_h\|_{1,\Omega}.$$